Asit Kumar Nayak

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Education

Indian Institute of Technology Madras **Jan 2023 – Present**

BS in Data Science and Applications

Relevant coursework: Machine Learning, Deep Learning and Generative AI, Business Analytics & Data Management, Statistics for Data Science, Mathematics for Data Science, Data Structures and Algorithms using Python, Database Management Systems, Software Engineering & Testing, Application Development, and Tools in Data Science.

Technical Skills

Programming languages: Python, SQL, Java, JavaScript, TypeScript.
Frameworks and libraries: scikit-learn, LightGBM, PyTorch, RDKit, NumPy, Pandas, Flask, FastAPI, React, Vue.js, SQLAlchemy, Jinja2.
Databases, platforms and tools: PostgreSQL, SQLite, Redis, Celery, Linux, Git, VS Code, Jupyter, Google Colab, Looker Studio.

Work Experience

Junior Developer Intern, NPTEL **Jan 2026 – Nov 2026**

Chennai, Tamil Nadu, India | On-site

- **Data pipeline automation:** Engineered automated pipelines to migrate and synchronize educational data across SWAYAM, TCS result systems, and the new NPTEL platform.
- **Business intelligence and analytics:** Collaborated with operations teams on Looker Studio dashboards and researched secure embedding methods for faculty and industry stakeholders.
- **Operational process improvement:** Gathered requirements for administrative editing actions and documented routine website operations to identify process-automation opportunities.
- **Program operations and data management:** Supported day-to-day NPTEL program operations, educational data management, and analysis of NPTEL datasets.

Teaching Assistant, Business Analytics, IIT Madras **May 2025 – Dec 2026**

Chennai, Tamil Nadu, India | On-site and Remote

- Assisted Prof. Malolan Sundararaman and Dr. Swaminathan with course delivery on data dashboarding, regression diagnostics, and logistic regression.
- Evaluated post-quiz and end-term assessments for a large learner cohort, supporting consistent grading and academic integrity.
- Demonstrated applied statistical methods through examples such as ANOVA for marketing impact analysis and regression for financial forecasting.

Teaching Assistant, Embedded C Lab, IIT Madras **May 2025 – Aug 2025**

Chennai, Tamil Nadu, India | On-site

- Guided students through embedded programming exercises in STM32CubeIDE (STM32 HAL) and Arduino IDE v2 (STM32duino core).
- Supported practical exercises in basic I/O, serial communication, external and timer interrupts, and sensor interfacing.

Teaching Assistant, Mentor and Content Developer **Oct 2023 – Sep 2024**

Mathematics for Data Science I and Möbius Platform | IIT Madras | On-site and Remote

- **Teaching and mentorship:** Supported university-level learners through course assistance and peer mentoring; delivered School Connect sessions for school students.
- **Mathematics and optimization:** Taught limits, continuity, derivatives, L'Hôpital's Rule, extrema, and function optimization; explained polynomial, quadratic, exponential, and logarithmic functions and their use in mathematical modelling.
- **Linear algebra and algorithms:** Covered coordinate geometry, straight-line fitting, matrix representations, linear systems, sequences and convergence; guided students through BFS/DFS, topological sorting, and shortest-path algorithms (Dijkstra's, Bellman–Ford, and Floyd–Warshall).
- **Möbius content development:** Created course materials and randomized practice questions for Mathematics for Data Science I with instructor Vishal.

Projects

System Threat Forecaster GitHub | Live Demo

Machine Learning Project

Developed a LightGBM-based binary classifier to estimate malware infection risk on Windows machines using antivirus telemetry and system-configuration data.

Music Genre Classifier GitHub | Live Demo

Deep Learning & Generative AI Project

Built a 10-class music genre classifier for noisy mashups using log-mel spectrograms and an ensemble of Audio Spectrogram Transformer (AST) and ResNet18 models.

SocietySync GitHub

Full-Stack Web Application

Developed a FastAPI and React platform for housing-society operations, including finance, resident helpdesk, amenity bookings, community updates, staff operations, and visitor management.

Hospital Management Application GitHub

Full-Stack Healthcare Application

Developed a role-based hospital management system for administrators, doctors, and patients, with appointment and treatment workflows using a Flask REST API, Vue frontend, and Celery/Redis background tasks.

Placement Portal Application GitHub

Full-Stack Web Application

Built a Flask-based campus placement portal for student and company profiles, placement drives, applications, shortlisting, and status tracking.

PresentationAI GitHub | Live Demo

AI-Powered Presentation Generator

Developed a multi-LLM presentation generator that converts text into PowerPoint slides and applies styles extracted from uploaded templates.

Hackathon

ANRF AISEHack 2.0 – Polymer Property Prediction GitHub | Competition | Email Proof

Team achievement: Top 20, Round 2 Private Leaderboard

Collaborated with teammates to develop models predicting seven polymer properties from SMILES using RDKit molecular features, multitask learning, scaffold-based validation, and ensemble methods.